

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 10

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Expand $(1 - 2x)^4$ and hence find the coefficient of x^2 in $(1 + 3x)(1 - 2x)^4$.

A 24

B -24

C 48

D 0

E 12

F -48

2 If $\log_a b = 2$ and $\log_b c = 3$, find $\log_a c$.

A 8

B 6

C $\frac{2}{3}$

D 9

E 5

F $\frac{3}{2}$

3 The interior angle of a regular polygon is 156° . How many sides does it have?

A 24

B 20

C 10

D 12

E 18

F 15

- 4 A geometric series has first term 2 and common ratio $r > 0$. The sum of the first three terms is 14. Find the sum of the first five terms.

- A 30
- B 52
- C 46
- D 94
- E 126
- F 62

5 The number $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$ is equal to

A 4

B $\frac{8}{\sqrt{2}}$

C $\sqrt{34}$

D 8

E $2\sqrt{2}$

F $4\sqrt{2}$

6 Evaluate $\int_{-1}^2 |x| dx$.

A 2

B $\frac{7}{2}$

C 3

D $\frac{5}{2}$

E $\frac{1}{2}$

F $\frac{3}{2}$

7 $(x + 2)$ is a factor of $2x^3 + ax^2 - 5x - 6$. Find the value of a .

A 5

B -2

C 1

D -3

E 2

F 3

- 8** A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?
- A** 6 cm
 - B** 12.5 cm
 - C** 8 cm
 - D** 7.2 cm
 - E** 5 cm
 - F** 6.4 cm

- 9** How many terms of the arithmetic series $3 + 7 + 11 + \dots$ are needed for the sum to first exceed 500?
- A** 17
 - B** 16
 - C** 11
 - D** 14
 - E** 18
 - F** 15

10 Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

A 36

B 240

C 0

D 84

E 120

F 156

11 Given that $\sin x + \cos x = 0$ and $0 \leq x < 2\pi$, find the sum of all possible values of x .

- A** π
- B** $\frac{5\pi}{2}$
- C** $\frac{7\pi}{4}$
- D** $\frac{3\pi}{4}$
- E** $\frac{3\pi}{2}$
- F** 2π

- 12** Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume 3.6 m^3 ? ($1 \text{ m}^3 = 1000 \text{ litres}$.)
- A** 30 minutes
 - B** 50 minutes
 - C** 5 minutes
 - D** 1 hour
 - E** 3 hours
 - F** 500 seconds

13 The graphs of $y = x^2 - 2x$ and $y = k$ meet at exactly two points. Find the complete set of values of k .

A $k > -1$

B $k \geq -1$

C $k > 0$

D $k = -1$

E $k < -1$

F all real k

14 The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

A 6

B 10

C 8

D 7

E 4

F 9

15 Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

A 1

B 2

C -1

D 5

E 3

F 0

- 16** Which of the following is the equation of the graph obtained by stretching $y = x^3 - 3x$ parallel to the x -axis with scale factor $\frac{1}{2}$?

A $y = \frac{1}{2}x^3 - \frac{3}{2}x$

B $y = 8x^3 - 3x$

C $y = 2x^3 - 6x$

D $y = \frac{x^3}{8} - \frac{3x}{2}$

E $y = 8x^3 - 6x$

17 When $x^3 - 3x^2 + 2x + 7$ is divided by $x^2 + 1$, what is the remainder?

A $x + 4$

B $x + 10$

C 10

D $-x + 10$

E $x + 7$

F $3x + 4$

18 Find the complete set of values of x for which $\frac{x-2}{x+1} \leq 0$.

A $-1 \leq x \leq 2$

B $x < -1$ or $x \geq 2$

C $-1 < x \leq 2$

D $x \leq 2$

E $x \leq -1$ or $x \geq 2$

F $-2 \leq x < 1$

- 19** $f(x)$ is a polynomial with $f'(x) = (x - 2)^2(x + 1)$. Which of the following is true?
- A** f has local minima at $x = -1$ and $x = 2$
 - B** f has no turning points
 - C** f has a local minimum at $x = -1$ and no turning point at $x = 2$
 - D** f has a local maximum at $x = -1$ and a local minimum at $x = 2$
 - E** f has a local minimum at $x = -1$ and a local maximum at $x = 2$

- 20 The expression $\frac{3}{7-2\sqrt{3}} - \frac{3}{7+2\sqrt{3}}$ simplifies to
- A $\frac{12\sqrt{3}}{37}$
- B $\frac{64}{6\sqrt{3}}$
- C $\frac{42}{37}$
- D $\frac{61}{42}$
- E 0
- F $\frac{12\sqrt{3}}{37}$

END OF TEST